

YASHAAS M

+91 7892343265

yashasteshi08@gmail.com

linkedin.com/yashaas

GitHub

PROFILE

Associate Engineer with hands-on experience in designing, developing, and deploying scalable backend systems and AI-driven applications across client and server environments. Experienced in debugging production issues, integrating new features, and working with modern development tools to deliver reliable and scalable software solutions within agile team settings.

TECHNICAL SKILLS

- Python
- RAG models
- Docker
- Java
- FastAPI
- LangGraph
- Chromadb
- Computer Vision
- Postgresdb
- LangChain
- Git and GitHub
- HTML/CSS/JS

EDUCATION

Bachelor of Engineering in Computer Science

CGPA :- 7.8

Channabasaveshwara Institute of Technology

2020 - 2024

Pre University Education

Percentage: 79

Seshadripuram Pre University College

2018- 2020

SSLC

Percentage: 94

Vidyaniketan High School

2017- 2018

PROJECTS

LLM-Based Multi-Agent Systems (AutoGen & LangGraph): Developed multiple AI agents using AutoGen, LangChain, and LangGraph integrated with LLM models via Groq API and Tavily for intelligent search. Built agents such as a Blog Writing Agent, Chess Playing Agent, and Weather Forecasting Agent with capabilities like context management, persistence, and multi-step reasoning, enabling efficient task automation and scalable AI-driven workflows.

WORK EXPERIENCE

RandomTrees Pvt. Ltd

Associate Engineer

Nov 2025 - Present

Chennai

Projects : ETRM (Oil and Gas Domain) [Terminal and BOL(Bill of Landing) Agents]

- Implemented a multi-agent RAG pipeline for the Terminal Management AI Agent using LangGraph, orchestrating three specialized nodes — an Orchestrator Node, an Intent Classifier Node, and a RAG Node — converting user queries into embeddings and performing semantic similarity search on ChromaDB collections for intelligent, context-aware response generation. Developed multi-session chat management with session-specific context tracking.
- Resolved SAS URL authentication token expiration issues in the BOL Agent by configuring longer-duration secure access.

BirlaNu : Automated Pipe Inventory Management Using Computer Vision

- Curated and annotated a custom pipe dataset using Roboflow, trained a custom Mask R-CNN model, and built a segmentation pipeline with IoU-based post-processing and elliptical crop extraction per detected segment.
- Conducted R&D on nested pipe ring counting evaluating Hough Circles, Contour Hierarchy, Radial Profile Analysis and SAM 3, integrated the final solution into a FastAPI backend with MaskR-CNN detection endpoints.
- Built and Dockerised a full-stack demo with a Streamlit UI featuring side-by-side comparison and detection metrics, deployed with Hugging Face Hub model delivery and PostgreSQL persistence.

CERTIFICATES

Mar - 2025

- Java Fullstack Developer :- [Certificate](#)

ACHIVEMENTS

- Developed job portal called [FreshJobHub](#).
- Developed a freelance [Personal Portfolio](#) website.